

Project Report: Diabetes Prediction using Naive Bayes and SVM

Mohamedsipli M
Roll Number: 216215006

Spring 2016

Submitted in fulfillment of the requirements for the degree of
Master of Science (M.Sc.) in Computer Science

Department of Computer Applications
National Institute of Technology, Tiruchirappalli

for the course
Data Mining (CAS764)

under the guidance of
Prof. Ramadoss

Abstract

This project investigates the application of machine learning models for the prediction of diabetes onset using the Pima Indians Diabetes Dataset. We implemented and evaluated three distinct classification approaches: Gaussian Naive Bayes, Support Vector Machine with Radial Basis Function (RBF) kernel, and Support Vector Machine with Linear kernel. The standard machine learning pipeline was followed, including data loading, crucial preprocessing steps such as handling missing values via median imputation, and feature scaling using StandardScaler. The data was split into training and testing sets to ensure robust evaluation. Model performance was assessed using key metrics including accuracy, confusion matrices, and Receiver Operating Characteristic (ROC) curves with Area Under the Curve (AUC). Visualizations were generated to provide insights into model performance and feature importance. On the test set, the SVM with RBF kernel achieved the highest accuracy (0.7403), while the Linear SVM demonstrated the highest AUC (0.81). This work, focusing on comparative analysis of these classifiers for a biomedical prediction task, was conducted as part of the Data Mining course (CAS764, Spring 2016) for the M.Sc. Computer Science program at the National Institute of Technology, Tiruchirappalli.

Contents

1	Introduction	3
2	Dataset	3
3	Methodology	3
3.1	Data Preprocessing	4
3.2	Data Splitting and Scaling	4
3.3	Model Training	4
3.4	Model Evaluation	4
4	Results and Discussion	5
4.1	Accuracy and Confusion Matrices	5
4.2	Feature Importance (Linear SVM)	5
4.3	ROC Curve Comparison	6
5	Conclusion	6

1 Introduction

Diabetes is a chronic metabolic disorder characterized by high blood sugar levels. Early prediction and diagnosis are crucial for effective management and preventing severe complications. Machine learning techniques offer powerful tools for analyzing medical data and building predictive models. This project explores the application of two fundamental machine learning classifiers, Naive Bayes and Support Vector Machines, to predict the likelihood of diabetes based on diagnostic measurements.

The primary goals of this project were:

- To load and preprocess the Pima Indians Diabetes Dataset, addressing common data quality issues.
- To implement and train Gaussian Naive Bayes and SVM models (with RBF and Linear kernels) for binary classification.
- To evaluate the performance of these models using appropriate metrics such as accuracy, confusion matrix, and ROC AUC.
- To visualize key results, including confusion matrices, feature importance, and ROC curves, to compare model effectiveness and gain insights into the data.

The project was conducted as part of the Data Mining course (CAS764, Spring 2016) in the M.Sc. Computer Science program at NIT Tiruchirappalli, under the guidance of Prof. Ramadoss.

2 Dataset

The dataset used in this project is the Pima Indians Diabetes Dataset, available from public repositories such as the UCI Machine Learning Repository. It contains medical predictor variables and one target variable, 'Outcome'. The predictor variables include:

- 'Pregnancies': Number of times pregnant
- 'Glucose': Plasma glucose concentration a 2 hours in an oral glucose tolerance test
- 'BloodPressure': Diastolic blood pressure (mm Hg)
- 'SkinThickness': Triceps skin fold thickness (mm)
- 'Insulin': 2-Hour serum insulin (μ U/ml)
- 'BMI': Body mass index ($\text{weight in kg}/(\text{height in m})^2$)
- 'DiabetesPedigreeFunction': Diabetes pedigree function
- 'Age': Age (years)

The 'Outcome' variable is 1 for diabetes positive and 0 for diabetes negative. The dataset contains 768 instances with 8 features and 1 target variable.

The data was loaded directly from the URL: <https://raw.githubusercontent.com/plotly/datasets/master/diabetes.csv>.

3 Methodology

The project followed a standard machine learning pipeline, as outlined in the following subsections.

3.1 Data Preprocessing

Initial inspection of the dataset revealed that certain columns ('Glucose', 'BloodPressure', 'SkinThickness', 'Insulin', 'BMI') contained '0' values, which are biologically implausible for these measurements and indicate missing data.

- These '0' values were explicitly replaced with 'NaN' (Not a Number) to be handled as missing data.
- Missing values ('NaN') in these columns were imputed using the median value of the respective column. The median was chosen for imputation as it is less sensitive to potential outliers compared to the mean.
- The dataset was then split into features (X) and the target variable (y).

3.2 Data Splitting and Scaling

- The dataset was split into training and testing sets with a ratio of 80% for training and 20% for testing using `sklearn.model_selection.train_test_split`. A `random_state` was set for reproducibility, and `stratify = y` was used to ensure that the proportion of the target class (diabetes positive/negative) was maintained in both the training and testing sets. The test set contained 154 samples.
- Feature scaling was performed using `sklearn.preprocessing.StandardScaler`. This transforms the features to have a mean of 0 and a standard deviation of 1. Scaling is particularly important for algorithms like SVM, which are sensitive to the magnitude and scale of features. The scaler was fitted on the training data and then used to transform both the training and testing data.

3.3 Model Training

Three different classification models were trained on the scaled training data:

- **Gaussian Naive Bayes:** A probabilistic classifier based on Bayes' theorem with the assumption of conditional independence between features. The Gaussian variant assumes that the continuous features associated with each class are distributed according to a Gaussian (normal) distribution.
- **Support Vector Machine (RBF Kernel):** SVM is a powerful algorithm that finds an optimal hyperplane to separate classes. The Radial Basis Function (RBF) kernel is a non-linear kernel that can handle complex decision boundaries.
- **Support Vector Machine (Linear Kernel):** An SVM model using a linear kernel, which finds a linear decision boundary. This model is also useful for interpreting feature importance based on the weights (coefficients) assigned to each feature.

The SVM models were initialized with `random_state=42` for reproducibility. The RBF SVM also had `probability=True` enabled to obtain probability estimates needed for the ROC curve.

3.4 Model Evaluation

Model performance was evaluated on the held-out test set using the following metrics:

- **Accuracy Score:** The proportion of correctly classified instances $((TP+TN)/(TP+TN+FP+FN))$.
- **Confusion Matrix:** A 2×2 table summarizing the number of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).
- **ROC Curve and AUC:** The Receiver Operating Characteristic (ROC) curve plots the True Positive Rate (Sensitivity) against the False Positive Rate (1 - Specificity) at various threshold settings. The Area Under the ROC Curve (AUC) provides a single measure of the model's ability to discriminate between the positive and negative classes, ranging from 0.5 (random guess) to 1.0 (perfect classifier).

4 Results and Discussion

After training the models and evaluating them on the test set, the following results were obtained.

4.1 Accuracy and Confusion Matrices

The accuracy scores for each model on the test set are summarized below:

- Naive Bayes Accuracy: **0.7013**
- SVM (RBF) Accuracy: **0.7403**
- SVM (Linear) Accuracy: **0.7013**

The confusion matrices provide a more detailed view of the models' performance. For the Linear SVM model, the confusion matrix on the test set was:

$$\begin{pmatrix} 82 & 18 \\ 28 & 26 \end{pmatrix}$$

In this matrix:

- TN (Top-Left): 82 instances were correctly predicted as negative (No Diabetes).
- FP (Top-Right): 18 instances were incorrectly predicted as positive (False Positive - Type I error).
- FN (Bottom-Left): 28 instances were incorrectly predicted as negative (False Negative - Type II error).
- TP (Bottom-Right): 26 instances were correctly predicted as positive (Diabetes).

Analyzing the confusion matrices for all models (including Naive Bayes and SVM RBF, whose matrices were also generated by the code) helps understand the types of errors each model makes. For diabetes prediction, False Negatives (failing to detect diabetes) are typically considered more costly than False Positives (incorrectly predicting diabetes).

4.2 Feature Importance (Linear SVM)

For the Linear SVM model, the coefficients learned by the model indicate the importance of each feature in determining the linear decision boundary. Features with larger absolute coefficient values have a greater influence. Based on the output, the feature importances (coefficients) sorted by absolute value are:

1. Glucose (0.9539)
2. BMI (0.5447)
3. Pregnancies (0.3343)
4. DiabetesPedigreeFunction (0.1724)
5. SkinThickness (-0.0647)
6. Age (0.0498)
7. BloodPressure (-0.0324)
8. Insulin (-0.0299)

The plot of these coefficients (generated by the code) clearly shows that Glucose and BMI are the most influential features in the linear SVM model, followed by Pregnancies. These findings align with medical understanding, as glucose levels and body mass index are strong indicators related to diabetes risk.

4.3 ROC Curve Comparison

The ROC curve comparison plot visualizes the trade-off between the True Positive Rate and False Positive Rate for each model across different thresholds. The Area Under the Curve (AUC) quantifies the overall performance. Based on the generated plot, the AUC values are:

- Naive Bayes AUC: **0.76**
- SVM (RBF) AUC: **0.80**
- SVM (Linear) AUC: **0.81**

Generally, a model with a higher AUC is considered better at distinguishing between the positive and negative classes. From these values, the SVM with the Linear kernel shows slightly better discriminative power than SVM RBF, and both SVM models perform better than the Naive Bayes model in terms of AUC on this test set. The ROC plot visually confirms how well each model's predictions separate the two classes.

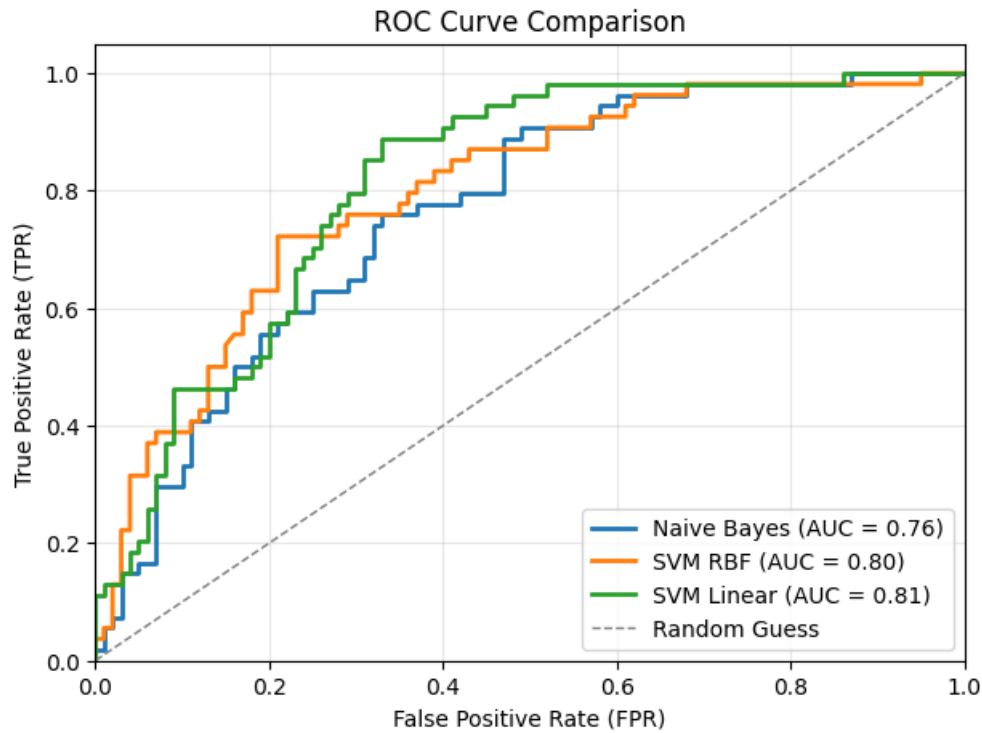


Figure 1: ROC Curve Comparison for Naive Bayes and SVM Models

5 Conclusion

This project successfully implemented and evaluated Gaussian Naive Bayes and Support Vector Machine models for diabetes prediction using the Pima Indians Diabetes Dataset. The preprocessing steps, including handling missing values and feature scaling, were crucial in preparing the data for modeling.

Based on the evaluation metrics (accuracy, confusion matrices, and ROC AUC), the SVM with the RBF kernel achieved the highest accuracy (0.7403) among the three models tested. The Naive Bayes and Linear SVM models showed identical accuracy (0.7013) on this test set, while the Linear SVM had the highest AUC (0.81), closely followed by SVM RBF (0.80). The analysis of the Linear SVM coefficients highlighted Glucose and BMI as the most influential features. The ROC curve comparison visually confirms the discriminative power of each model.

While the models achieved reasonable accuracies, limitations exist. The dataset size is relatively small, which can impact model generalization. The imputation method used is basic; more sophisticated techniques could be explored. The models used are standard classifiers; trying ensemble methods or deep learning could yield different results.

For future work, exploring more advanced preprocessing techniques, performing hyperparameter tuning using cross-validation, investigating other machine learning algorithms, and potentially incorporating domain knowledge for feature engineering could further improve the prediction accuracy and robustness of the model.

References

- Smith, J.W., Everhart, J.E., Dickson, W.C., Knowler, W.C., Bennett, P.H. (1988). Using the ADAP learning algorithm to forecast the onset of diabetes mellitus. In Proceedings of the Annual Symposium on Computer Application in Medical Care (pp. 261-265). (Source of the dataset)
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., ... Duchesnay, E. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825-2830.